

MUHAMMED SHAHIQ A R

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Professional Summary

AI Engineer and AI & Data Science graduate with practical experience developing LLM applications, RAG systems, Computer Vision workflows, and full-stack ML pipelines. Proven ability to translate business problems into engineering solutions and ship secure, scalable, production-grade AI products.

Technical Skills

Machine Learning & AI	TensorFlow, PyTorch, scikit-learn, Keras
Data Processing	NumPy, Pandas, Seaborn
Deep Learning / CV / NLP	CNN, RNN, Transformers, YOLO, OpenCV, spaCy, NLTK
LLMs / Generative AI	GPT, LLaMA, Hugging Face Transformers, LangChain
Deployment	Docker, FastAPI
Programming	Python, SQL, R

Key Projects

Mental Health RAG Chatbot 2025

Python, LangChain, Gemini 2.0 Flash, ChromaDB

- Designed a context-aware RAG chatbot using **LangChain and ChromaDB**, enabling multi-turn conversation support with controlled retrieval behavior.
- Implemented **safety guardrails and prompt-injection prevention logic** to ensure secure, on-topic responses in sensitive user interactions.
- Developed data ingestion and embedding pipelines in **Python**, improving retrieval consistency and response relevance.
- Integrated **sliding-window memory** to maintain short-term conversational context and reduce hallucination risk.

Healthcare Voicemail Triage System 2024

Python, OpenSMILE, Scikit-learn, Flask, NLP

- Built an end-to-end **voicemail triage pipeline** combining acoustic features and **NLP-based** text signals for urgency assessment.
- Developed a **hybrid classification workflow** using **Scikit-learn** models and rule-based logic to improve prioritization accuracy.
- Implemented **PII-redaction and text-sanitization logic** to support privacy-aware processing of patient information.
- Designed data preprocessing and inference pipelines in **Python**, supporting iterative retraining and improvement.

Internship Experience

Zoftcares Solutions LLP Tirur, Kerala

Artificial Intelligence (R&D) Intern Feb 2025 – Aug 2025

- Designed and launched **Trusity**, a student-focused LLM-powered RAG chatbot that converts ideas into real-world business concepts, incorporating strong input/output guardrails to prevent unsafe or sensitive responses.
- Engineered safe-chat workflows using **LangChain, RAG pipelines**, and prompt-leak prevention techniques to ensure secure, reliable, and context-aware AI interactions.
- Conducted research and experimentation with GenAI tools to design **scalable, production-ready retrieval and response architectures**.

- Contributed to **AITIQS (AI-Powered Incident Management System)** by improving detection accuracy by **30%** and supporting deployment and optimization on **AWS**.
- Deployed ML and AI services using **FastAPI** and **Docker**, delivering robust, scalable APIs for real-time, production use.
- Retrained and iteratively improved models with new data to enhance **adaptability, accuracy, and rule-based detection performance**.

Education

MES College of Engineering
B.Tech in Artificial Intelligence and Data Science

Kuttippuram, Kerala
2021 – 2025

- **Relevant Coursework:** Deep Learning, Machine Learning, Data Science, Research Methodology

Government Higher Secondary School, NGO Quarters
Higher Secondary Education (Class 12) — PCMCS

Kerala
2019 – 2021

- Developed strong **communication, teamwork, and collaboration** skills through academic and group activities.